

Demand-Supply Theory, General Equilibrium Theory, and Macroeconomics as Special Cases

A Reclassification of Mainstream Economics under the Survival-Breakdown Model

Toyoharu Higuchi

Independent Researcher

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Note on the English Version

The authoritative original version of this paper is the Japanese-language original.

This English version was prepared with AI-assisted translation support because the author is a native speaker of Japanese and not a native English speaker. Every effort has been made to preserve the meaning and structure of the original, but in the event of any discrepancy, the Japanese original shall prevail.

This English version is provided for reference only. It should be treated as supplementary material accompanying the original Japanese paper.

Abstract

This paper argues that demand-supply theory, general equilibrium theory, and macroeconomics are not foundational first principles of economics, but lower-order and conditional special-case theories that acquire theoretical meaning only after more prior economic conditions have already been secured. Standard economics usually treats demand and supply, price adjustment, equilibrium, and aggregate consistency as the starting points of analysis. Yet this theoretical order must be regarded as explanatorily too late. Before those concepts can operate, agents must first be constitutionally established, remain within the survivable region, retain basic feasibility, and be able to participate in markets in a sustainable manner. If these prior conditions are impaired, demand and supply, equilibrium, and macroeconomic regularities cannot function as general theoretical starting points.

This paper does not aim to restate the foundational theory of the Survival-Breakdown Model (SBM) itself. The theoretical order of constitutive conditions, survivability, basic feasibility, continuation, and then optimization has already been presented in the published SBM foundational paper. The task here is, on the basis of that foundational theory, to reposition demand-supply theory, general equilibrium theory, and macroeconomics within the theoretical order of SBM and to systematically redescribe their theoretical status.

On the basis of the SBM research program, this paper reorders three central domains of mainstream economics within one continuous explanatory sequence. First, demand and supply are reinterpreted as local states of market adjustment that have meaning only when constitutive conditions, survivability, basic feasibility, continuation, and sustainable participation have already been secured. Second, general equilibrium is reclassified as a special case of systemic adjustment defined only over a selected participant domain that already satisfies those conditions. Third, macroeconomics is repositioned as a field that should ask first about the distribution of continuation, destabilization, threshold proximity, and the possibility of cascading breakdown across households, firms, and institutions, rather than taking aggregate consistency itself as the starting point.

The contribution of this paper is not to deny these theories, but to reclassify them. Demand-supply analysis remains useful within relatively stable domains. General equilibrium analysis can retain analytical value in domains where participation and feasibility have been secured in advance. Macroeconomics is also valid where the structure of continuation has not been substantially eroded. But none of them should be treated as the universal starting point of economics. What economics must ask first is not where prices move, whether equilibrium exists, or whether aggregate consistency is maintained, but whether agents, organizations, and

market participant populations remain within a domain that is operable at all. Only after that prior condition is satisfied do these concepts acquire limited validity.

Keywords

Survival-Breakdown Model (SBM); demand and supply; general equilibrium; macroeconomics; market participation; sustainability; basic feasibility; continuation; aggregate instability; prior economic conditions

JEL Classification

B41, B49, D01, D40, D50, E10

Chapter 1. Introduction

Demand-supply theory, general equilibrium theory, and macroeconomics have long been treated as core theories of modern economics. Demand-supply theory analyzes the adjustment of prices and quantities in individual markets, general equilibrium theory deals with the consistency of the economy as a whole across multiple markets, and macroeconomics takes aggregate fluctuations such as output, employment, prices, investment, and policy effects as its object. Although these three theories differ in their objects of analysis and levels of abstraction, they have all played central roles in understanding economic order.

However, the fact that these theories are all important is not the same as saying that they are first principles of economic theory. The question of this paper is whether concepts such as demand, supply, equilibrium, and aggregate consistency should be placed at the beginning of theory, or whether they are derivative concepts that acquire meaning only after more prior conditions have been satisfied. The claim here is not that existing theories should be invalidated. Rather, their theoretical status and explanatory order must be rearranged.

This paper does not aim to restate the foundational theory of the Survival-Breakdown Model (SBM) itself. The basic structure of SBM - namely, the theoretical order of constitutive conditions, survivability, basic feasibility, continuation/destabilization/termination, and then optimization - has already been presented in the published foundational paper. The task here is to show, on the basis of that foundational theory, how demand-supply theory, general equilibrium theory, and macroeconomics should be repositioned within it.

In standard economics, it is often presupposed that households and firms are already constituted, already participate in markets as choice-capable agents, respond to prices, express demand and supply, and generate equilibrium and aggregate outcomes as the result of their adjustment. This construction is analytically powerful and remains highly effective in many stable situations. But its effectiveness does not automatically guarantee its validity as the starting point of theory. Demand, supply, equilibrium, and aggregate quantities all presuppose that the agents and organizations carrying them are themselves surviving and remain inside a domain in which continuation is possible.

If economic agents are to be theorized, what must first be confirmed is whether the relevant agent is constitutionally established, whether it remains within the survivable region, whether it possesses a basically feasible range of action, whether it is continuable through time, and whether it can participate in markets in a sustainable way. If these conditions are not met, then even if price responses, demand functions, supply functions, equilibrium conditions, or aggregate equations are defined, they still lack the prior conditions that

support the theory at its root. The issue, therefore, is not the truth or falsity of particular propositions inside existing theories, but the order in which explanation begins.

In this respect, the basic stance of this paper is clear. Demand-supply theory, general equilibrium theory, and macroeconomics have each treated themselves as first principles under different implicit formulations of the starting point of economics. Yet they are in fact lower-order theories at different levels, all resting on the same prior conditions. Demand-supply theory deals with local market adjustment in situations where agents are already able to participate in markets. General equilibrium theory deals with systemic adjustment in situations where such agents can exist simultaneously across multiple markets. Macroeconomics deals with aggregate adjustment in situations where those continuable structures are still sufficiently maintained at the level of the economy as a whole.

What becomes clear through this repositioning is that the difficulty with these three theories lies, prior to their particular content, in how their explanations begin. Before asking where prices move, there must already exist agents capable of responding to prices. Before asking whether equilibrium exists, there must already be a maintained population of agents able to participate across markets. Before asking about aggregate consistency or policy effects, households, firms, and institutions must themselves remain continuable and not already be dissolving through chains of breakdown.

The purpose of this paper is to demonstrate this point systematically on the basis of the architecture of SBM. What matters here is not abolishing existing theories, nor merely adding one more constraint. What SBM calls into question is not the behavioral constraints of already constituted agents, but the prior conditions that determine whether such agents exist in theory at all. In this sense, this paper is not a paper that negates existing theories, but one that redescribes their conditions of validity, their effective range, and their theoretical status.

From this standpoint, the central claim of this paper can be summarized as follows. Demand-supply theory is a special case of local market adjustment, general equilibrium theory is a special case of systemic adjustment, and macroeconomics is a special case of aggregate adjustment. And all of them operate only when prior conditions - constitution, survivability, basic feasibility, continuation, and sustainable participation - have already been satisfied. Accordingly, the beginning of economics must be placed not in demand and supply, nor in general equilibrium, nor in macroeconomics, but in the foundation that makes them possible.

The structure of this paper is as follows. Chapter 2 formulates, as a general principle grounded in the theoretical order of SBM, why none of demand-supply theory, general equilibrium theory, and macroeconomics can be a first principle. Chapter 3 reinterprets demand and supply as a theory of local market adjustment that has meaning only over sets of agents filtered by prior conditions. Chapter 4 reclassifies general

equilibrium as a special case of systemic adjustment defined only over a selected participant domain. Chapter 5 repositions macroeconomics as a theory that should begin not from aggregate consistency itself, but from the distribution of continuation and destabilization. Chapter 6 clarifies the linkage among the three levels. Chapter 7 explains why the point is reclassification rather than abolition. Chapter 8 concludes.

Chapter 2. General Principle - Why Demand-Supply, Equilibrium, and Aggregation Are Not First Principles

The task of this chapter is to show, not as separate objections but as a general principle, why none of demand-supply theory, general equilibrium theory, or macroeconomics can count as a first principle of economic theory. The issue here is not the truth or falsity of the internal propositions of each theory. The problem is that all of them begin by presupposing an economy in which agents are already constituted and already active. Demand, supply, equilibrium, and aggregate consistency are important concepts for explaining the economy. But they are not concepts that should be placed at the beginning of theory.

In the foundational theory of SBM presupposed here, economic agency is understood in the order of constitutive conditions, survivability, basic feasibility, continuation, and only then optimization. The importance of this order is not merely that it rearranges topics. It establishes an explanatory priority in the sense that earlier stages provide the conditions of possibility for later ones. If an agent is not constitutionally established, survivability cannot even be meaningfully asked. If the agent does not remain within the survivable region, basic feasibility as a meaningful range of action cannot be secured. And if continuation is not maintained, optimization cannot function as the guiding structure of action.

The most concise way to express this order is the minimal form of survival and breakdown. When an agent has available resources and there is a minimum amount required for continued existence, the first question of theory is not which option the agent prefers, but whether the agent is inside the continuable region at all. In this context, the survival condition is not an exogenous extra constraint added to a standard optimization problem. It is the threshold that determines, prior to lower-order concepts such as preference ordering, utility comparison, demand response, and supply response, whether the agent has theoretical meaning at all.

In this sense, the position of this paper is not to define mainstream theory as simply false, but to revise its explanatory order. These theories still retain major analytical power within stabilized domains. Prices can mediate adjustment, equilibrium conditions can describe consistency, and macroeconomic variables can serve as objects of policy analysis. Yet they do not explain first how agents are constituted, how they are maintained, or how they can fall into destabilized states. What existing theory basically treats is an economy that is already in operation.

When this general principle is applied to demand-supply theory, it becomes clear that neither demand nor supply is foundational to economics. Demand is an expression of discretionary choice that has meaning only when constitutive conditions, survivability, and basic feasibility have already been secured. Supply is an expression of productive and market activity that has meaning only when continued operation, payment capacity, access to inputs, and participation capacity are maintained. Accordingly, the market must be theorized first not as a place of price-clearing, but as a place in which sustainable participation is already established.

Applying the same principle to general equilibrium theory shows that even if general equilibrium represents a general structure of simultaneous adjustment across many markets, it is still not a first principle. General equilibrium theory usually asks about the consistency of price vectors given agents, endowments, technology, and market access. From the SBM perspective, however, being a nominal participant and being an effective participant must be distinguished. Only after the conditions of constitution, survivability, basic feasibility, continuation, and sustainable participation are satisfied does an agent belong to the domain over which equilibrium is meaningful.

It is further appropriate to think in the same order in macroeconomics. Standard macroeconomics usually puts aggregation, output, employment, prices, investment, shocks, and stabilization at the center of analysis. Yet before such aggregate consistency has meaning, one must ask whether households, firms, and institutions remain inside the continuation domain, how much instability has accumulated, how threshold proximity is distributed, and how deterioration can propagate. The proper object of macroeconomics is therefore not aggregate consistency itself, but instability inside an interconnected structure of continuation.

The implication that follows is clear. Demand-supply theory, general equilibrium theory, and macroeconomics are not three independent foundational theories. They belong to different levels - local market adjustment, systemic adjustment, and aggregate adjustment - but all depend on the same foundational layer. Demand-supply theory occupies the layer of local market adjustment, general equilibrium theory the layer of its economy-wide extension, and macroeconomics the layer of aggregate expression. Yet all three presuppose the prior conditions of constitution, survivability, basic feasibility, continuation, and sustainable participation.

Accordingly, the general principle established in this chapter can be summarized as follows. First, the starting point of economic theory must be placed not in optimization, price adjustment, equilibrium, or aggregate consistency, but in the conditions of the constitution and continuation of agency. Second, demand-supply theory, general equilibrium theory, and macroeconomics should all be understood as conditional and special-case theories that operate only after those prior conditions have already been satisfied. Third, the significance of existing theories is not denied, but their theoretical status is changed. They are not universal starting points, but lower-order theories supported by a prior foundation.

Chapter 3. Demand and Supply Are Special Cases

In standard economics, the relation between demand and supply has been treated as the most elementary and most general starting point of market theory. Households generate demand, firms generate supply, and prices adjust the two. This schema has been widely accepted as a foundational explanatory structure of economics. Yet in light of the theoretical order established in the previous chapter, this construction begins too late. Before demand and supply can stand in a systematic relation, both the demand side and the supply side must first be constituted as economic agents, have survivability secured, remain in safe conditions, possess basic feasibility, and stay within a domain in which continuation and sustainable market participation are possible. Accordingly, demand and supply are in principle independent and unrelated, and the demand-supply relation should be understood not as a universal first principle of economics but as a lower-order and conditional theory of local market adjustment that operates only after prior conditions have already been secured.

3.1 Why Standard Demand-Supply Theory Begins Too Late

Standard demand-supply theory looks elementary because it starts from a simplified model of market life that is already functioning. On one side stands the consumer who chooses goods according to preferences; on the other stands the firm that supplies in response to prices and costs. The market is then depicted as a place where these two curves meet and are adjusted by price. But this schema tacitly presupposes agents who are already constituted, firms that are already active, and the prior possibility of market participation.

From the perspective of SBM, however, this order is explanatorily too late. A household as a demand-bearing agent must first satisfy constitutive conditions. The basic conditions of life must be in place, and minimal bodily and social safety must be secured. When the conditions sustaining the capacity to choose goods are undermined, the issue is not which goods are preferable, but whether the status of the agent as a choosing subject is itself being maintained. Further, if survival security is not secured, demand does not appear in the ordinary sense as a normal act of choice. Mere survival is still not enough. What is also required is a minimum margin for choice, namely basic feasibility. Even when an agent remains above the survival threshold, if payment capacity, mobility, time margin, or information access are severely compressed, ordinary demand no longer takes shape.

The same applies on the supply side. Standard theory assumes that a firm already in operation adjusts the quantity supplied in response to price. But for a firm to be a supplier at all, the continuation of the firm must first be maintained. If assets, expense payments, wage payments, procurement, or the ability to place goods on the market have broken down, the first problem of the firm is not the choice of supply quantity, but whether it

can continue operating without falling into suspension. Supply, too, is therefore a lower-order concept that exists only on the basis of continuation.

The market itself is no different. Even where an institutional place of exchange exists, if participation capacity has been impaired the market exists only nominally. Access to the market, procedural conditions, time margin, access to information, and payment capacity may all deteriorate to the point that one cannot actually participate in a market that formally exists as an institution. Hence the market relation that demand-supply theory has treated as a starting point in fact rests on more basic conditions of agency constitution and participation. The problem with standard demand-supply theory is not abstraction as such, but that it begins by treating as already satisfied the conditions that should have been asked about first.

3.2 Demand Is a Lower-Order Concept

Given the order above, demand is not the first expression of economic agency. Demand is a lower-order expression that appears only on top of already stabilized agency. Only when an agent satisfies constitutive conditions, survives, remains in a safe condition, and retains basic feasibility does demand choice in the usual sense acquire theoretical meaning. Demand is therefore not a primitive concept directly derived from utility maximization, but a conditional behavioral expression within a domain in which more basic conditions have already been met.

In minimal form, let the set of actions available to the agent be A , the execution cost of those actions be $c(a)$, and the available basic margin be B . Then the basically feasible action set can be written as

$$A^{\text{feas}}(B) := \{a \in A \mid c(a) \leq B\} \quad (3.1)$$

For demand to be defined in the ordinary sense, this set must not be empty. That is,

$$A^{\text{feas}}(B) \neq \emptyset \quad (3.2)$$

must hold. Only when this condition is satisfied can the agent choose within the feasible set. Accordingly, ordinary choice is represented in the form

$$\max_{a \in A^{\text{feas}}(B)} U(a) \quad (3.3)$$

but what matters here is not optimization itself, but the prior fact that $A^{\{feas\}}(B)$ is nonempty. Demand is a concept that should be theorized only after constitutive conditions, survivability, and basic feasibility have already been secured. If the agent has not reached that level, then apparent demand behavior should not be understood as movement along an ordinary demand curve, but as pressure on the constitution of agency and the compression of basic margin.

3.3 Supply Is Also a Lower-Order Concept

Supply, too, is not a primitive concept of economics. It is a lower-order concept that has meaning only when a continuable productive agent still maintains operations and the chain of inputs, payments, management, and transactions is still functioning. Standard theory assumes that the firm already exists, is already operating, and then adjusts supply in response to price. From the SBM perspective, however, this order too must be reversed. For a firm to be a supplier, it is first necessary that continuation capacity has not been impaired and that the firm is not already moving toward suspension or functional breakdown.

In this sense, supply is not merely a response of output quantity, but a conditional response of firms that remain inside the continuation domain. Even if a firm solves a profit-maximization problem, that has theoretical meaning only when continuation is already secured. In situations where the continuation of the firm is collapsing, profit maximization is not the primary problem. The primary problem is the preservation of continuation conditions such as ongoing payments, employment maintenance, asset retention, credit maintenance, and operation maintenance. Accordingly, the supply curve is not the most basic theoretical expression of the firm. It is only a local expression valid where a continuable firm is still able to carry out market responses. Standard theory treats supply as primitive only by pushing the more basic conditions of the existence and continuation of firms outside theory.

3.4 The Market Must First Be Defined as a Site of Sustainable Participation

The market itself should not be positioned from the outset as a mechanism of adjustment. It must first exist as a site in which sustainable participation is possible. Even where a market exists institutionally, if mobility, time margin, access to information, procedures of market participation, or payment capacity are impaired, participation remains merely nominal. Accordingly, the minimal filter for market participation can be written as

$$P^{\min} = \{i \mid V_i \geq 0, R_i \geq 0\} \quad (3.4)$$

where V_i denotes participation capacity and R_i denotes basic feasibility. Demand and supply acquire theoretical meaning only over the set of agents that pass through such a participation filter. The market is therefore first a site of sustainable participation and only second a site of price adjustment. The order is not the

reverse. This minimal filter is a simplified expression focusing on participation capacity and basic feasibility; below it will be extended into a fuller participation condition that also includes constitutive conditions, survivability, and continuation capacity.

Let the nominal sets of households and firms, together with their respective effective subsets, be written as

$$H^* = \{h \in H \mid C_h = 1, S_h = 1, R_h \geq 0, K_h \geq 0, V_h \geq 0\} \quad (3.5)$$

$$F^* = \{f \in F \mid C_f = 1, S_f = 1, R_f \geq 0, K_f \geq 0, V_f \geq 0\} \quad (3.6)$$

Then the only agents that have theoretical meaning as demand-side and supply-side agents are those belonging respectively to these effective subsets. Here C denotes constitutive conditions, S survivability, R basic feasibility, K continuation capacity, and V participation capacity. Demand and supply should therefore be theorized not as the behavior of nominal agents in general, but only as conditional responses of agents that have passed through these five prior conditions.

3.5 Demand-Supply Functions Have Meaning Only over a Filtered Domain

If this point is written in terms of market response, the demand response of household h and the supply response of firm f are respectively defined as

$$\tilde{x}_h(p), \tilde{y}_f(p) \quad (3.7)$$

Here the tilde indicates that these are not responses of the entire nominal population of agents, but responses defined only over the set of agents filtered by prior conditions. If the effective participation domain in market m is written as

$$\Omega_m^* = \{(h, f) \mid h \in H^*, f \in F^*, \text{ and market } m \text{ satisfies the conditions for participation in the market}\}$$

then market demand and market supply are

$$D_m(p) = \sum_{h \in H_m^*} \tilde{x}_{hm}(p) \quad (3.9)$$

$$S_m(p) = \sum_{f \in F_m^*} \tilde{y}_{fm}(p) \quad (3.10)$$

Accordingly, the demand-supply relation does not hold from the outset as a universal relation over the entire nominal market. Local adjustment through prices has theoretical meaning only within a participant domain where constitutive conditions, survivability, basic feasibility, continuation capacity, and participation capacity have already been secured to some degree. This does not make the standard demand-supply diagram entirely invalid. Rather, it clarifies its proper status. It is not a universal first principle, but a schematic of local adjustment valid only within a stabilized domain of market participation - that is, a diagram of a special case.

The conclusion of this chapter is therefore clear. Neither demand, nor supply, nor the market, nor price adjustment is the first object of economic theory. What must first be asked is whether agents and firms are constituted at all, whether they are continuable, and whether they are able to participate. Demand and supply are in principle unrelated; only when the prior condition for their relation is exceptionally met does the demand-supply relation acquire meaning as a theory of local market adjustment. Demand-supply theory should not be abolished, but its theoretical status must be changed. It is not a universal first principle of economics, but a special case supported by prior conditions.

Chapter 4. General Equilibrium Is a Special Case

General equilibrium theory has been regarded as one of the most sophisticated systematic constructions in modern economics, as a theory that coordinates multiple markets simultaneously. If demand-supply theory deals with local adjustment in a single market, general equilibrium theory extends this to multiple markets and asks about the consistency of the economy as a whole. Yet in light of the argument up to the previous chapter, this systematic extension too begins too late as an explanation. Before consistency across multiple markets can have meaning, the households and firms participating in those markets must themselves first be effective participants who satisfy constitutive conditions, survivability, basic feasibility, continuation capacity, and participation capacity. If demand and supply are already lower-order conditional concepts, then general equilibrium as their multi-market consistency is even more clearly a lower-order conditional concept. General equilibrium theory should therefore be reclassified not as a first principle of economics, but as a theory of systemic adjustment that holds only over a participant domain that has passed through prior conditions.

4.1 Why Standard General Equilibrium Theory Begins Too Late

Standard general equilibrium theory begins from households already defined, consumption sets already meaningful, production sets already meaningful, endowments already available, and markets already operating. Households are given as choosing agents, firms as producing agents, market participation is assumed possible, and exchange is assumed executable. The question then becomes whether markets are jointly consistent under a price vector. Formally, this construction is powerful. Yet from the perspective of SBM it begins from a domain that has already been constituted without explaining how that domain itself comes into being.

A household becomes a meaningful demander only when constitutive conditions are maintained, survivability is secured, basic feasibility is retained, continuation is possible, and market participation itself is sustainable. A firm likewise becomes a meaningful supplier only when continuation capacity is preserved, operation is substantively maintained, and participation capacity still survives. Moreover, general equilibrium deals not with a single market but with the multi-market consistency of such populations. Therefore, if even the demand-supply relation in a single market already presupposes prior conditions, then general equilibrium as the systematic totality of such relations presupposes them all the more deeply. The issue is not that general equilibrium theory is abstract. The issue is that it treats the very domain of equilibrium as if it were given, without explaining how that domain is constituted.

In this sense, general equilibrium theory is not always wrong. Within relatively stable domains it can be analytically useful. But its usefulness does not imply theoretical priority. General equilibrium is a subsequent theory that presupposes the existence of an already stabilized, participable domain. The problem with general equilibrium theory is therefore one of misclassification. What is problematic is that it has been treated as the

universal analytical starting point of the economy. Repositioned instead as a conditional theory of systemic adjustment, its analytical value can be retained.

4.2 Nominal Participant Sets and Effective Participant Sets

This point can first be clarified by making explicit the nominal participant set. Let the nominal participant set be

$$N = \{1, 2, \dots, n\} \quad (4.1)$$

This nominal set N merely represents the total set of agents institutionally or formally located within the economy, and by itself does not give the proper domain of equilibrium analysis. The set that has analytical meaning for equilibrium analysis is instead the effective participant set that has passed through more prior conditions. It can be written as

$$P = \{i \in N \mid C_i = 1, S_i = 1, R_i \geq 0, K_i \geq 0, V_i \geq 0\} \quad (4.2)$$

where C denotes constitutive conditions, S survivability, R basic feasibility, K continuation capacity, and V participation capacity. Thus, the participant set that general equilibrium theory has tacitly assumed is not theoretically primitive as it stands. Equilibrium has meaning only over the domain of participants selected in this way.

If this integrated expression is written separately for the set of households H and the set of firms F , letting the nominal household set be H and the nominal firm set be F , then their respective effective subsets are

$$H^* = \{h \in H \mid C_h = 1, S_h = 1, R_h \geq 0, K_h \geq 0, V_h \geq 0\} \quad (4.3)$$

$$F^* = \{f \in F \mid C_f = 1, S_f = 1, R_f \geq 0, K_f \geq 0, V_f \geq 0\} \quad (4.4)$$

That is, households and firms belong to the domain of the equilibrium problem only after satisfying the five prior conditions of constitutive conditions, survivability, basic feasibility, continuation capacity, and participation capacity. In what follows, the equilibrium problem is considered over and .

4.3 Choice Sets and Production Sets Are Also Filtered by Prior Conditions

The choice sets faced by households and the production sets faced by firms are not theoretically valid in their nominal form as such. Nominal choice sets and nominal production sets must be understood as effective

sets that have passed through prior conditions. Thus, if the effective choice set of households and the effective production set of firms are written as

$$X_h^* \subseteq X_h, Y_f^* \subseteq Y_f \quad (4.5)$$

then the object of equilibrium analysis does not lie from the outset on the nominal choice set or the nominal production set as such, but only on these selected sets. Since demand and supply were already conditional lower-order concepts, general equilibrium as their multi-market extension also has meaning only over a selected domain. What matters here is not simply that the sets are technically narrowed. More importantly, the object of theory itself changes. What general equilibrium theory ought to treat is not the behavior of nominal participants, but the consistency structure of agents whose possibility of constitution and continuation has already been secured.

4.4 Equilibrium Conditions Have Meaning Only over the Participant Domain

Accordingly, equilibrium conditions too should be defined not over nominal sets, but over the population of agents that has passed through prior conditions. What general equilibrium theory treats is not the totality of agents that exist legally or nominally, but the multi-market consistency that obtains among effective participants who retain constitutive conditions, survivability, basic feasibility, continuation capacity, and participation capacity. Hence the essence of the equilibrium condition lies not so much in the algebraic form itself as in the set over which that form is defined. Using the effective participant domain defined up to this point, the equilibrium condition should be rewritten as a consistency condition over selected participants.

To make the continuity with Chapter 3 explicit, let us adopt one notational convention. In Chapter 3, local market responses defined over selected agent sets were denoted with tildes. In this chapter, to rewrite the same responses as objects of multi-market consistency in the context of general equilibrium, we denote them with stars. The star therefore does not introduce a new population of agents; it simply rewrites the selected responses introduced in Chapter 3 over the domain of general equilibrium. In this sense, the response functions in this chapter are a systematic extension of the responses in Chapter 3.

In what follows, let $x_h^*(p)$ denote the net demand vector expressed by household h under price vector p given endowments, and let $y_f^*(p)$ denote the effective production-supply plan vector of firm f . Thus, on the household side, $x_h^*(p)$ is described not as gross demand including endowments but as net demand after deducting endowments. On the firm side, $y_f^*(p)$ is described as the effective production-supply plan expressed by firms that still retain continuation and participation capacity. Under this convention, the effective responses of households and firms are written as

$$x_h^*(p), y_f^*(p) (h \in H^*, f \in F^*) \quad (4.6)$$

What matters here is that these responses are not defined over nominal agents in general, but only over the selected participant set N^* that has already passed through prior conditions.

In that case, the effective excess demand of the economy as a whole is obtained by subtracting the sum of firms' effective production-supply plans from the sum of households' net demands, and is written as

$$Z^*(p) = \sum_{h \in H^*} x_h^*(p) - \sum_{f \in F^*} y_f^*(p) \quad (4.7)$$

Here $Z^*(p)$ is not excess demand for the totality of nominal participants, but excess demand restricted to the effective participant domain. Accordingly, even if this expression is formally similar to the standard excess-demand equation in general equilibrium theory, its domain of definition is different. What matters from the standpoint of SBM is not the algebraic shape of the equilibrium condition, but over what participant domain that shape has meaning.

The general equilibrium condition is written, when there exists a price vector p^* at which this effective excess demand becomes zero, as

$$Z^*(p^*) = 0 \quad (4.8)$$

This does not mean that all nominal markets are automatically cleared. Rather, it means that the market responses defined over the population that still retains constitution, survival, feasibility, continuation, and participation are consistent under a certain price vector. Equilibrium must therefore be understood not as the original state of the economy as a whole, but as conditional consistency within a selected participant domain.

To write this by market m , let the set of households and the set of firms that satisfy effective participation conditions in that market be H_m^* and F_m^* respectively. That is, H_m^* is the set of households within H^* that can participate effectively in market m , and F_m^* is the set of firms within F^* that can participate effectively in market m . Then the market-specific equilibrium condition is

$$\sum_{h \in H_m^*} x_{hm}^*(p) = \sum_{f \in F_m^*} y_{fm}^*(p) \quad (4.9)$$

Here the left-hand side is the sum of the net demands of effective households in market , and the right-hand side is the sum of the production-supply plans of effective firms in market . Market equilibrium, too, therefore has meaning not as a condition of institutional markets in general, but only where there still remains a population of agents able to participate effectively in that market.

What is important here is not to deny general equilibrium theory. On the contrary, the point is to locate its valid scope more rigorously. General equilibrium theory can remain useful as a description of multi-market consistency so long as an effective participant domain is still maintained. But it is not a first principle of economics. Before the equilibrium condition, one must ask who remains within the domain of the equilibrium problem at all. General equilibrium theory is therefore a lower-order special-case theory that operates only over a participant domain that has passed through prior conditions.

Chapter 5. Macroeconomics Is a Special Case

Macroeconomics has remained one of the central domains of economics as a theory dealing with aggregate relations such as output, employment, investment, prices, and policy effects. Yet from the perspective of SBM, macroeconomics too is not a first principle of economic theory. Standard macro theory often takes aggregate consistency, equilibrium paths, steady states, and adjustment from shocks as the starting points of analysis. But before such aggregate consistency has meaning, the distribution of households, firms, and institutions must itself still remain inside the domain of continuation. If that prior condition is missing, aggregate relations themselves lose their standard theoretical status.

5.1 Why Standard Macroeconomic Theory Begins Too Late

Standard macro theory treats households as ordinary agents of consumption and labor supply, firms as ordinary agents of production and employment, and institutions as ordinary mechanisms of policy transmission. But this schema too presupposes an aggregate operative domain that is already in place. It implicitly treats as given the conditions under which households can be consumers, firms can be operating producers, and institutions can still connect effectively to the relevant population.

From the perspective of SBM, however, the first macroeconomic issue is not the equilibrium or fluctuation of aggregate variables. What must be asked earlier is whether households, firms, and institutions still remain inside a domain of continuation. In situations where households approach the survival threshold, firms lose continuation capacity, and institutions shrink in their capacity to connect, aggregate consistency itself is already being eroded. The issue then is not the size of deviation from equilibrium, but whether equilibrium and consistent aggregate relations are still theoretically operable at all.

Accordingly, the problem with standard macro theory is not that it treats aggregates as such. The problem is that it begins by treating as already satisfied the continuation conditions that should have been asked about first. Macroeconomics cannot be the first description of the economy as a whole. It is a subsequent aggregate theory that stands on top of a continuation structure that has already been stabilized to some degree.

5.2 The Foundation of Macroeconomics Lies in the Distribution of Continuation States

This point can be expressed in a minimal dynamic form by letting the continuation balance of each agent n be

$$B_{t+1}^n = B_t^n + u_t^n - c_n \quad (5.1)$$

where u_t^n denotes inflows such as income, support, recovery, and maintenance, and c_n denotes the minimum required cost of continuation. The significance of this equation is that it treats the agent first not as an optimizing subject, but as an entity governed by the dynamics of continuability. The foundation of macroeconomics must therefore be placed not in aggregate production functions or aggregate demand curves, but in the distribution of such continuation states.

What matters here is that B_t^n is not limited to a merely monetary balance. For households it may be interpreted as the margin for maintaining life; for firms, as the margin for continuing operations; and for institutions, as the margin for execution and connection. This minimal dynamic therefore becomes a basic form for describing continuation, destabilization, and termination in a unified way across types of agents.

5.3 Classification into Continuation, Destabilization, and Termination

The dynamics of the continuation balance describe the state of each agent through a threshold structure. If θ_n denotes the destabilization threshold for each agent, then the state can be classified as

$$B_t^n > \theta_n \Rightarrow \text{continuation} \quad (5.2)$$

$$0 < B_t^n \leq \theta_n \Rightarrow \text{destabilization} \quad (5.3)$$

$$B_t^n \leq 0 \Rightarrow \text{termination / functional breakdown} \quad (5.4)$$

What matters here is that destabilization is placed at the center of theory rather than treated as a peripheral exception. In standard theory such states are often handled as special cases of frictions or shocks. But in the order of SBM, aggregate consistency itself is instead a later phenomenon that appears only when the distribution of such states is sufficiently contained. The center of macroeconomics is therefore not 'deviation from equilibrium,' but the erosion of the continuation domain and its distribution.

5.4 Introducing Aggregate Vulnerability

As a minimal indicator expressing this distribution in aggregate form, aggregate vulnerability can be defined as

$$\Phi_t = \frac{\sum_{n \in H \cup F \cup I} w_n \mathbf{1}(B_t^n \leq \theta_n)}{\sum_{n \in H \cup F \cup I} w_n} \quad (5.5)$$

Here H, F, and I are respectively the sets of households, firms, and institutions, w_n is the weight assigned to agent n, and $\mathbf{1}(\cdot)$ is the indicator function. Thus Φ_t gives the weighted proportion of agents that are at or

below the destabilized state. In particular, when the weights w_n are normalized to 1, the denominator can be omitted.

Its theoretical significance is clear. What macroeconomics ought to ask first is not GDP gaps or deviations from equilibrium paths, but the spread of such unstable agents. Aggregate regularities themselves presuppose that this unstable distribution is suppressed to a certain degree. Thus Φ_t is a minimal aggregate indicator that shifts the starting point of macroeconomics away from aggregate consistency itself and toward the degree of erosion of the continuation structure.

5.5 Macroeconomic Dynamics with Propagation

Further, the evolution of continuation balances depends not only on each agent's own inflows and costs, but also on deterioration and support transmitted through interdependence. A minimal extension is written as

$$B_{t+1}^n = B_t^n + u_t^n - c_n + \phi_n(\text{support, recovery}) - \psi_n(\text{shocks, deterioration of linkages}) \quad (5.6)$$

where ϕ_n represents the effect of support or recovery, and ψ_n the effect of shocks or deterioration in interconnection. In this way, instability is no longer merely an internal problem of each agent, but can propagate cumulatively among households, firms, and institutions. The basic object of macroeconomics is therefore not an equilibrium with frictions added to it, but the process through which erosion of the continuation domain is transformed into aggregate instability.

The key point here is to understand macro-instability not as a temporary response to an exogenous shock, but as a process in which declines in continuation capacity expand distributionally and relationally. Destabilization of one household can depress firm revenues through reduced consumption; destabilization of firms can lower the of other agents through reduced employment and investment; and deterioration in connection to institutions can weaken by reducing the reach of support. Thus, the core of macroeconomics lies not in the static consistency of aggregates, but in the distribution and propagation of continuation balances.

5.6 Reclassifying the Status of Macroeconomics

Accordingly, the theoretical status of macroeconomics is clearly repositioned. Macroeconomics remains useful in dealing with aggregate variables such as output, employment, investment, and policy effects. But these are not the first objects of economic theory. What must be asked first is whether households, firms, and institutions remain inside the continuation domain, how far destabilization has accumulated, and how widely the possibility of cascading breakdown has spread. Macroeconomics should therefore be understood not as the first theory of aggregate adjustment, but as a lower-order conditional theory of aggregate adjustment that becomes meaningful only after a continuable structure has already been secured.

Chapter 6. Linking the Three Layers - Local Adjustment, Systemic Adjustment, and Aggregate Adjustment

Up to the previous chapter, it has been shown that demand-supply theory, general equilibrium theory, and macroeconomics are not independent first principles, but theoretical domains that operate only after more prior conditions have been secured. The task of this chapter is not merely to reclassify the three separately, but to make explicit the order in which they are linked. What matters is that the three are not unrelated theories, but a lower-order sequence built on top of the same foundational layer.

Demand-supply theory, treated in Chapter 3, deals with price responses and quantity adjustment in local markets. It presupposes that demand-side and supply-side agents are already able to participate in markets. The theory therefore has meaning only when constitution, survivability, basic feasibility, continuation, and sustainable participation have already been secured. Demand-supply theory remains important under those conditions, but its object is limited to local market adjustment.

General equilibrium theory, treated in Chapter 4, extends this local market adjustment to multiple markets. What general equilibrium theory asks is simultaneous consistency across markets and the coherence of the price system. But this adjustment problem has meaning only when effective participants are constituted in each market. Since markets themselves already depend on prior conditions, the market system taken as their totality cannot escape the same dependence. General equilibrium theory is therefore not a first principle, but a higher-order systematic extension built on local adjustment.

Macroeconomics, treated in Chapter 5, takes aggregate fluctuations such as output, employment, investment, prices, and policy effects as its object. Yet these aggregate variables too presuppose that households, firms, and institutions still remain within the continuation domain and that destabilization or termination has not become generalized. The object of macroeconomics is therefore not an independent starting point disconnected from local or systemic adjustment. It is a theory that asks what aggregate consequences are generated by the interdependence of agents that already stand on top of prior conditions.

Seen in this way, the relation among the three is clear. Demand-supply theory deals with local adjustment, general equilibrium theory with systemic adjustment, and macroeconomics with aggregate adjustment. These are different analytical levels, but all share the same feature of beginning from an economy that is already in operation. Precisely because of that commonality, all three are lower-order theories located above the foundational layer. What economics must ask first is not how prices move, whether equilibrium exists, or how

aggregate quantities fluctuate, but whether the conditions of the constitution and continuation of agents are in place at all.

Accordingly, the reclassification proposed here does not separately demote three theories one by one. Rather, it clarifies the order among them and reorganizes their theoretical status. Above the foundational layer lies demand-supply theory as local adjustment; above that lies general equilibrium theory as systemic adjustment; and above that lies macroeconomics as aggregate adjustment. Through this order, the three central components of mainstream economics are understood not as multiple first principles that mutually support one another, but as a stratified lower-order series resting on a common foundation.

The conclusion of this chapter is therefore simple. Demand-supply theory, general equilibrium theory, and macroeconomics belong to different analytical levels - local adjustment, systemic adjustment, and aggregate adjustment - but all are lower-order theories that stand above the foundational layer. The starting point of economics must therefore be placed in none of the three, but in the conditions of agency constitution and continuation that make them possible.

Chapter 7. Not Abolition but Reclassification

The argument so far has shown that demand-supply theory, general equilibrium theory, and macroeconomics are not first principles, but theories that have meaning only after prior conditions have been secured. Yet this conclusion is easy to misunderstand. The position of this paper is not to declare these theories invalid. What is at issue is not their analytical usefulness itself, but their status as the universal starting point of theory.

This distinction is important. Demand-supply theory remains useful in relatively stable market domains. General equilibrium theory also retains value as a formal framework for examining systemic consistency where participant sets and market roles are effectively maintained. Macroeconomics likewise remains effective for analyzing aggregate fluctuations and policy effects in situations where the continuation structure has not been seriously impaired. Thus, the claim of this paper is not to discard the achievements of existing theory. So long as their conditions of application are satisfied, existing theories remain valid within their own domains.

What this paper changes is their location. In the standard understanding, demand-supply theory, general equilibrium theory, and macroeconomics are easily juxtaposed as central theories representing different aspects of economics. From the perspective of SBM, however, they are not parallel first principles. All presuppose that agents are constitutionally established, survivable, basically feasible, continuable, and able to participate sustainably. They are therefore not foundational theories, but conditional theories that operate only on top of the foundational layer.

Nor does this paper dispute whether poverty, minimum survival, breakdown, or instability can be formally represented within standard theory. Even if such representability exists, it does not by itself confer explanatory priority. The issue is what should count as the first question of theory. What SBM emphasizes is that the conditions of the constitution and continuation of agency precede price adjustment, equilibrium, and aggregate consistency. In this sense, what this paper proposes is not the addition of a new auxiliary constraint, but a revision of theoretical order itself.

Accordingly, the expression 'not abolition but reclassification' most accurately captures the conclusion of this paper. Demand-supply theory has a clear role as a theory of local adjustment, general equilibrium theory as a theory of systemic adjustment, and macroeconomics as a theory of aggregate adjustment. But none of them marks the beginning of economics. The beginning of economics must be placed in the foundational layer that makes those theories meaningful - namely, the conditions of the constitution and continuation of agency. Existing theories are not negated; rather, their conditions of validity and their effective range are delimited and their status is reorganized.

The conclusion is therefore clear. The three core theories of mainstream economics are not to be maintained as universal starting structures, but reclassified as conditional and lower-order theories located above the foundational layer. What is carried out here is not the abolition of theory, but the reordering of theory and the repositioning of theoretical status.

Chapter 8. Conclusion

This paper has provided a unified reclassification of three central domains - demand-supply theory, general equilibrium theory, and macroeconomics - from the perspective of the Survival-Breakdown Model (SBM). The conclusion is clear. None of these three is a foundational first principle of economics. Demand-supply theory is a theory of local market adjustment when agents are already able to participate in markets; general equilibrium theory is a theory of systemic adjustment when such participants can be simultaneously coordinated across multiple markets; and macroeconomics is a theory of aggregate adjustment when the continuation structure remains sufficiently maintained at the level of the economy as a whole.

The central point shown here is not that the three core theories of mainstream economics are false. The problem lies in the theoretical order in which they have been placed. Price adjustment, equilibrium, and aggregate consistency are important for describing the economy. But before they acquire meaning, agents, organizations, and market participant populations must already be constitutionally established, remain inside the survivable region, be basically feasible, be continuable, and be able to participate in a sustainable way. Accordingly, what economic theory must ask first is not where prices move, whether equilibrium exists, or how aggregate variables behave, but whether the domain in which such questions are meaningful exists at all.

In this sense, the contribution of this paper lies not in abolishing existing theory, but in redescribing its theoretical status. Demand-supply theory can retain limited validity as a theory of local market adjustment, general equilibrium theory as a theory of systemic adjustment, and macroeconomics as a theory of aggregate adjustment. Yet none of them provides the conditions for the constitution and continuation of agency. The significance of SBM is not simply to add a separate higher theory in place of mainstream theory, but to clarify where mainstream theory holds, where it ceases to hold, and in what order it should be positioned.

The theoretical order implied by this paper is therefore not the usual juxtaposition of demand-supply theory -> general equilibrium -> macroeconomics. Rather, there is first a foundational layer consisting of constitutive conditions, survivability, basic feasibility, continuation, and sustainable participation, and above that are located demand-supply theory as local market adjustment, general equilibrium theory as systemic adjustment, and macroeconomics as aggregate adjustment. Through this repositioning, the three central concepts of mainstream economics are no longer treated as universal starting points placed at the head of economic theory, but as lower-order special cases that have meaning only above the foundational layer.

In this sense, the paper has presented a minimal theoretical framework for positioning demand-supply theory, general equilibrium theory, and macroeconomics as unified special cases under SBM. The tasks opened by this paper are clear. First, the major objections anticipated against this reclassification should be examined more precisely as problems of theoretical order. Second, SBM should be developed not only as a reordering

of demand, equilibrium, and macroeconomics, but as a broader research program. Yet within the scope of this paper at least, one point is clear: demand-supply theory, general equilibrium theory, and macroeconomics are not the first beginning of economics, but lower-order theories that acquire meaning only under prior conditions.

References

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